

FC423 - Injector Timing Rail 1 Fuel Pressure - Data Erratic, Intermittent, or Incorrect

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check for multiple fault codes.	
	STEP 1A. Read the fault codes.	Fault Codes 113 and 114 active?
STEP 2.	Measure fuel inlet restriction.	
	STEP 2A. Measure restriction.	Fuel restriction within specification?
STEP 3.	Check for a mechanical failure of the pump.	
	STEP 3A. Check the sleeve current.	Sleeve current changing?
STEP 4.	Clear the fault codes.	
	STEP 4A. Disable the fault code.	Fault Code 423 inactive?
	STEP 4B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check for multiple fault codes.

STEP 1A. Read the fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Read the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Fault Codes 113 and 114 active? YES	Appropriate troubleshooting tree
	Fault Codes 113 and 114 active? NO	2A

STEP 2. Measure fuel inlet restriction.

STEP 2A. Measure restriction.

Conditions :

- Turn keyswitch ON.
- Start the engine and let it idle.

Action	Specification/Repair	Next Step
Measure the fuel inlet restriction. • Refer to Procedure 006-020 (Fuel Inlet Restriction) in Section 6 of the Service Manual, QST30 Series Engines, Bulletin 4021539. (/qs3/pubsys2/xml/en/procedures/57/57-006-020.html)	Fuel inlet restriction within specification? YES	3A
	Fuel inlet restriction within specification? NO Repair : Refer to Procedure 006-020 (Fuel Inlet Restriction) in Section 6 of the Service Manual, QST30 Series Engines, Bulletin 4021539. (/qs3/pubsys2/xml/en/procedures/57/57-006-020.html)	Appropriate troubleshooting tree

STEP 3. Check for mechanical failure of the pump.

STEP 3A. Check the sleeve current.

Conditions :

- Turn keyswitch ON.
- Start the engine and let it idle.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check the sleeve current. • Use INSITE™ electronic service tool to monitor the sleeve current while varying the engine rpm.	Sleeve current changing? YES Repair : Repair or replace the fuel pump. • Refer to Procedure 005-016 (Fuel Pump) in Section 5 of the Service Manual, QST30 Series Engines, Bulletin 4021539. (/qs3/pubsys2/xml/en/procedures/57/57-005-016.html)	4A
	Sleeve current changing? NO Repair : • Call for pre-authorization. • Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/60/60-019-031.html)	4A

STEP 4. Clear the fault codes.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to verify that Fault Code 423 is inactive.	Fault Code 423 inactive? YES	4B
	Fault Code 423 inactive? NO Repair : Verify that all steps have been completed. If all steps have been completed, then follow the technical escalation process.	1A

STEP 4B. Clear the inactive fault codes.

Conditions :

- Connect all components
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO Repair : Troubleshoot any remaining fault codes.	Appropriate troubleshooting charts

Last Modified: 12-Dec-2016
